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Real-Time Crack Detection on Concrete Using Non-Destructive Approaches Based on Yolo Series: Review

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Abstract. Concrete is a fundamental material in infrastructure often exhibits significant surface cracking over time due to exposure to harsh environmental conditions. Cracking is considered one of the most critical forms of deterioration. In recent years, numerous disasters globally have underscored this vulnerability. Visual sensing and deep learning has marked a significant evolution, offering a leap from the manual, expert-dependent methods to a more continuous, data-driven approach. This improvement facilitated the integration of new components for structural health monitoring such YOLO series. It is characterized by their affordability, high resolution, and reliability. This paper aims to provide a short review of the current state of deep learning methodologies in real-time crack detection, with a particular focus on the application of the YOLO series. It presents also the main key components in the YOLO platform, the progresses and updates in each release. Despite the significant advancements to enhance detection capacities, several challenges remain. Summary of challenges and recommendations will be underlined at the end of this article.

Keywords. Crack detection, deep learning, prediction, AI, Yolo series

1. Introduction

Infrastructure components such as buildings, power plants, roads, and other commonly utilized structures play a critical role in the daily lives of citizens. These elements are indispensable to the sustained development and economic prosperity of contemporary societies. Given their importance, these structures warrant heightened attention towards sustainability to ensure longevity and resilience. Particularly, concrete -- a fundamental material in infrastructure often exhibits significant surface cracking over time due to exposure to harsh environmental conditions and human-induced stressors. Concrete, while not as strong or tough as steel, boasts many beneficial properties, including resistance to fire, water, corrosion, and cyclic loading. These characteristics make it a

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popular material for a wide range of infrastructure such as bridges, pavements, tunnels, dams, foundations, and various building components like beams, columns, and walls. However, concrete's low tensile strength makes it particularly susceptible to cracking, which is considered one of the most critical forms of deterioration. In recent years, numerous disasters globally have underscored this vulnerability.

These incidents highlight the need for ongoing infrastructure monitoring to safeguard economic development and public safety [1, 2]. Traditional inspections by experts are time-consuming, labour-intensive, and expensive. In this context, the advent of visual sensing technologies and deep learning (DL) has marked a significant evolution, which offers a leap from the manual, expert-dependent methods to a more continuous, objective, and data-driven approach. These technologies, leveraging advancements in digital imaging, have made vision-based methods increasingly popular for crack detection and structural health inspection, integral components of structural health monitoring (SHM). The advanced methods are affordable, high resolution, and reliable, suggesting that the inspection process could be divided into five principal stages: acquisition, detection, measurement, assessment (or analysis), and decision. The early identification of cracks and structural anomalies has contributed to the collection of data on a continuous basis, using the visual systems to give important information on maintenance planning. The designing of automatic algorithms for crack detection, segmentation, and classification would ensure accurate and reliable infrastructure monitoring. During the last decade, indeed, a plethora of automatic algorithms in this area have been developed, and great enhancements were seen in recent years, mostly due to the rise of DL and AI techniques [2–4].

DL and AI (figure 1) have remarkable potential to develop non-destructive methods in the inspection of concrete structures. These technologies are geared not only toward early detection of structural anomalies but also toward scalability and automation of monitoring systems. In this case, DL emerges as an apparent revolutionary technology for detecting cracks using artificial neural networks for learning and extracting meaningful features from the data received by the vision system. The models can be trained on large datasets that, in fact, give them the possibility to recognize and classify cracks with a surprisingly high level of accuracy. The You Only Look Once (YOLO) series [5], particularly effective for real-time object detection, has gained considerable attention for its applicability to crack detection.

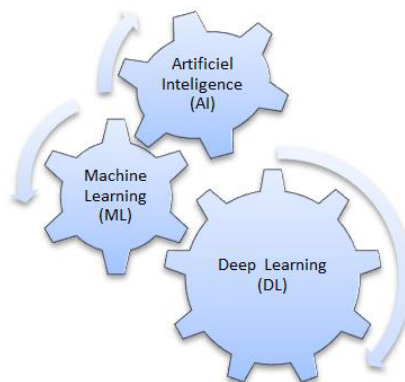


Figure 1. DL is the driving concept for the ML which is one of the AI approaches [4].

The use of the vision-based approaches, DL, and advanced segmentation and classification of cracks enables the derivation of detailed inferences about the crack patterns, sizes, orientations, and severity. This places good knowledge and basis for engineers and maintenance people to clearly identify and prioritize the maintenance activities to have critical problems resolved as soon as possible, while less severe cracks can be taken care of on an appropriate schedule.

This study provides a short review of the state of the art in deep learning methodologies for real-time crack detection, with a special focus on members of the YOLO series. The review further delves into the YOLO architecture, its evolution through various versions, and adaptability for specific demands for real-time crack detection in concrete structures. By modifications and improvements in the YOLO framework, attempts have been made to optimize accuracy, speed, and reliability, which are further highlighted by selected case studies and implementations presented in this study. This paper gives more attention to AI techniques integrated to better improve detection performance and future perspectives.

2. YOLO Series in Real Time Concrete Crack Detection

YOLO is a popular series of DL models for real-time object detection. Among some of the key features that make it effective for crack detection are architecture, performance, and accuracy:

(a) Architecture: It operates on a single-stage neural network that performs both object classification and location prediction in one step, making the approach much faster than traditional two-stage methods.

(b) Performance: It does process images at a rate of real-time applications, which is crucial in field inspections.

(c) Accuracy: It achieves good accuracy in detecting large or small features objections.

High-resolution images of concrete surfaces with well labelled cracks, were strongly required for the network, which could intensely help the model analyse accuracy. As well, some data augmentation methods could be used, such as rotation, scaling, and flipping, which contribute to the creation of training datasets. The Yolo series has been upgraded for 10 generations since its initial release in 2016, with YOLOv10 being the latest, released at the beginning of 2024. Over the past eight years, researchers have dedicated efforts to address the shortcomings of the original Yolo framework, particularly its difficulty in detecting small-sized objects. The recent versions include backbone networks, data augmentation strategies, and optimization techniques which help to improve the speed and the accuracy while maintaining real-time capabilities. One fact is that Yolo requires to fine-tune pre-trained model on concrete crack datasets to adjust the model weights. Strategies such as transfer learning can help to reduce the training time and data requirements. In term of implementation, the model can be integrated into various platforms, including mobile and web applications or deployed on drones and other inspection device after adequate training. It can be combined with other non-destructive testing methods (such as thermography) to confirm the detection results. The main key components in YOLO Architectures are:

(1) The convolutional layers: Used for extracting features from the input image.

(2) The down sampling layers: Used to reduce the spatial resolution of feature maps.

(3) **The batch normalization:** Used to improve the training speed and the stability.
 (4) **The activation functions:** Rectified Linear Unit (ReLU) for intermediate layers and sigmoid for bounding box confidence and class probabilities.

(5) **The residual connections:** In later versions, residual connections help in training deeper networks without degradation.

The flow chart process (figure 2) starts by taken an image as input and resized to a predetermined size. This image is resized and divided into small grids. Each grid cell will be responsible for predicting objects whose centres fall within that cell. Then, for each grid cell, the network predicts a certain number of bounding boxes along with confidence scores for each box indicating the likelihood that an object is present and its precision. The class probabilities for the detections are also predicted by the grid cell. Then, we apply a non-maximum suppression to eliminate redundant bounding boxes based on their confidence scores and intersection over Union values. The result is a refined set of bounding boxes with the highest confidence scores for detected objects. The final output consists of bounding boxes with class labels, confidence scores and their respective probabilities, ready for further processing or visualization

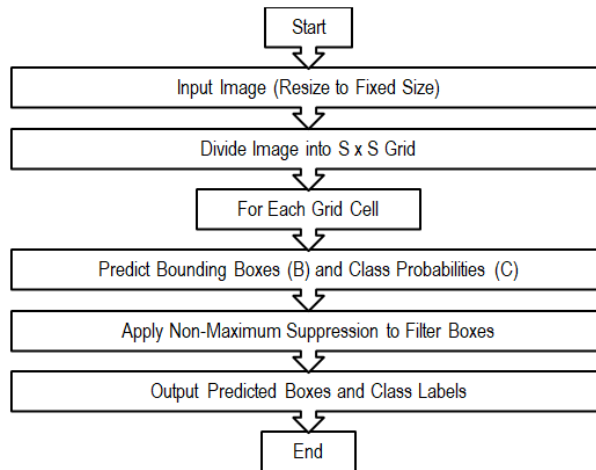


Figure 2. Flow chart process for YOLO.

Since 2020 with the introduction of YOLOv4, It marked a leap forward with the introduction of a CSP backbone, spatial pyramid pooling, path aggregation networks, and enhanced data augmentation techniques. Two years later, Wu et al. [6] combine a pruning technique with the EvoNorm-S0 structure under YOLOv4-based network to achieve a 1.5% increase in accuracy to reach 90% in concrete crack detection. Due to its detection speed (measured in frames per second - FPS), Zhang et al. [7] introduced an automated bridge surface crack detection and segmentation algorithm that leverages an improved YOLOv4 structure to accurately identify cracks and their approximate locations from multi-object images. Yu et al. [8], their modified network achieved real-time processing speeds of 119 FPS with images at the thousand-pixel level, setting a new standard for detection speed that is particularly beneficial for real-time applications in civil engineering. Despite these advancements, challenges such as high false-negative rates in specific applications persist.

The release of YOLOv5 marked another rapid evolution in the series, addressing complex real-world challenges such as pavement crack recognition. It was applied successfully to detect objects in milliseconds [9]. Due to these advantages, YOLOv5 has inspired a surge of modifications across thousands of algorithms. It introduced some significant modules into the structure, such as the C3-faster module, SPPF module, and Ghostconv module. Liang et al. [10] developed an improved model designed to detect defects in large metal stampings, focusing on enhanced feature extraction and incorporating a multi-headed self-attentive mechanism to effectively identify wrinkles, holes, and cracks. One of the YOLO series' distinct advantages is the flexibility it offers to make targeted modifications to parts of the algorithm such as the backbone, neck, and head while retaining the integrity of other components. Zhao and his associates [11] proposed RDD-YOLO mode based on Res2Net blocks as a new backbone, along with modifications to the neck and heads. Later, Li et al. [12] conducted a comparative study on the performance of modified versions 4 and 5 for real-time instance-level detection on asphalt pavement. An average precision of 99.4% was demonstrated across different versions including v3, v4, and v5, highlighted the efficiency and detection capabilities of these models. However, the efficiency still rely heavily on extensive training data, which can lead to overfitting. This dependence not only challenges the balance between performance, computational efficiency, and accuracy but also increases the need for substantial GPU resources and meticulous data annotation.

YOLOv6 is a more specific generation in the YOLO series. It is a significant refinement of its predecessors, focusing on enhancing the renowned speed and accuracy, particularly in real-time object detection. However, this version is tailored for commercial use, which limit its applicability in research.

Wang et al. [13] introduced later YOLOv7, which significantly improved detection speeds, boosting FPS from 5 to 120. While the impact of YOLOv7 has been clear, driving significant interest and inspiring numerous research initiatives, the implementation of YOLOv7 in specific applications, such as rail defect detection using B-scan images, poses unique challenges. The current research of YOLOv7 exposes reasons for missing detection of smaller defects and inadequate attention to subtle injuries. To address these issues, Li et al. [14] incorporated the Convolutional Block Attention Module (CBAM) into the YOLOv7 architecture, enhancing recall and accuracy for B-scan image detection. The architecture of YOLOv7 optimized the object detection problem by fusing down-sampling and up-sampling layers in a balanced manner, essentially getting to the improvement of feature extraction for different details with respect to the object scale. Throughout the optimizations, YOLOv7 is effective for real-time applications like concrete crack detection, combining speed with precision in object detection. It has been pivotal in advancing concrete crack detection technology, integrating swin-transformer blocks [15] and various algorithmic enhancements [16] so as to improve detection performance. Some modifications were achieved by introducing an initial convolutional layer to extract basic features, followed by several convolutional block subsampling (CBS) layers after extracting basic features by initial convolutional layer, which contributed to decrease spatial resolution while expanding feature map depth. This process ensured detailed feature capture with reduced computational cost. Enhanced Layer Aggregations (ELAN) ensured comprehensive feature maps for accurate detection by integrating different-level features. Up-sampling, facilitated by transposed convolutions, refines resolution, aiding in precise object localization. Jump connections within YOLOv7 link early and deep layers, merging

low and high-level features for improved accuracy. Predefined anchor boxes within the network help predict object positions and sizes, supported by classification and regression layers that specify categories and bounding coordinates.

YOLOv8 [17] builds on YOLOv7's capabilities, focusing on speed and adaptability for diverse environments. This latest model optimizes its architecture with A Robust Backbone, Advanced Head Network and Sophisticated Loss Optimization. A recent innovation includes the YOLOv8-GAM-Wise-IoU model, specifically developed for automated bridge surface crack detection. This model integrates a global attention module (GAM), a wise intersection over union (IoU) loss function, and a decoupled head, significantly improving performance metrics [18]. YOLOv9 evolves previous architecture, offering refinements that aim to streamline detection processes further and improve accuracy, especially in real-time applications.

3. Future Challenges and Perspectives

Despite the significant advancements in the YOLO model to enhance detection capabilities, several challenges and research gaps remain that require further exploration to optimize performance and applicability. Below some challenges and recommendations:

Integration of advanced attention mechanisms: While recent modifications to the YOLO architecture have incorporated attention mechanisms to enhance feature extraction and fusion, the application of these mechanisms often introduces complexity and may impact real-time processing capabilities. Challenge lies in integrating more sophisticated attention mechanisms that do not substantially increase the computational cost. Focus could be on attention models that reduce computational cost while still improving the accuracy and robustness of the detection system.

Environmental conditions: The accuracy of defect detection can dramatically deteriorate when influenced by changing conditions as mentioned above. Modelling and developing adaptive mechanisms are important in handling the adaptation of models in various environmental conditions. There is a potential direction for further research is to invest in more multi-condition training systems or to develop an algorithm in which can autonomously adjust parameters based on the environmental conditions presented to the model.

Real-time problem solving: Part of the challenge of detection can be that it should be able to trigger in real-time, for example being able to convincingly detect a crack on an edge of a structure. However, it does not only have to start fast but be able to run on some hardware with the model requesting an unseen set of resources. Methods such as altering architectures or any other strategies to focus on performance or reducing latency as an attempt to real-time detection.

Overfitting: Models will often overfit in any DL system, but this is an ongoing consideration in model development for DL for concrete crack detection. For operational real-world environments, there are additional regularisation techniques that can be investigational to consider for further projects. Further, DL models that are evaluated on a dataset that includes labelled test set(s) that is not arbitrary from the training dataset, designed to expose models trained to be safe in a real-world environment. Would help close the loop from improvement to research to practice.

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References

- [1] Yan Z, Teng S, Luo W, Bassir D. Bridge modal parameter identification from UAV measurement based on empirical mode decomposition and fourier transform. *Appl. Sci.* 2022; 12: 8689.
- [2] Teng S, Chen X, Chen G, Cheng L and Bassir D. Structural damage detection based on convolutional neural networks and population of bridges. *Measurement.* 2022; 202: 111747. doi: 10.1016/j.measurement.2022.111747.
- [3] Teng S, Chen G, Yan Z, Cheng L and Bassir D. Vibration-based structural damage detection using 1-D convolutional neural network and transfer learning. *Structural Health Monitoring.* 2023; 22(4): 2888–2909.
- [4] Bassir D, Lodge H, Chang H, Majak J and Chen G. Application of artificial intelligence and machine learning for BIM. *International Journal for Simulation and Multidisciplinary Design Optimization.* 2023; 14(5).
- [5] Sapkota KR. YOLOv10 to its genesis: A decadal and comprehensive review of the you only look once (YOLO) series. 2024. <https://arxiv.org/pdf/2406.19407>.
- [6] Wu P, Liu A, Fu J, Ye Y and Zhao Y. Autonomous surface crack identification of concrete structures based on an improved one-stage object detection algorithm. *Engineering Structures.* 2022; 272: 114962. doi: 10.1016/j.engstruct.2022.114962.
- [7] Zhang J, Qian S and Tan C. Automated bridge surface crack detection and segmentation using computer vision-based deep learning model. *Engineering Applications of Artificial Intelligence.* 2022; 115: 105225. doi: 10.1016/j.engappai.2022.105225.
- [8] Yu Z, Shen Y and Shen C. A real-time detection approach for bridge cracks based on YOLOv4-FPM. *Automation in Construction.* 2021; 122: 103514. doi: 10.1016/j.autcon.2020.103514.
- [9] Xu G, Yue Q and Liu X. Deep learning algorithm for real-time automatic crack detection, segmentation, qualification. *Engineering Applications of Artificial Intelligence.* 2023; 126: 107085. doi: 10.1016/j.engappai.2023.107085.
- [10] Liang Y, Feng S, Zhang Y, Xue F, Shen F and Guo J. A stable diffusion enhanced YOLOV5 model for metal stamped part defect detection based on improved network structure. *Journal of Manufacturing Processes.* 2024; 111: 21–31. doi: 10.1016/j.jmapro.2023.12.064.
- [11] Zhao C, Shu X, Yan X, Zuo X and Zhu F. RDD-YOLO: A modified YOLO for detection of steel surface defects. *Measurement.* 2023; 214: 112776. doi: 10.1016/j.measurement.2023.112776.
- [12] Li J, Yuan C and Wang X. Real-time instance-level detection of asphalt pavement distress combining space-to-depth (SPD) YOLO and omni-scale network (OSNet). *Automation in Construction.* 2023; 155: 105062. doi: 10.1016/j.autcon.2023.105062.
- [13] Wang CY, Bochkovskiy A and Liao HM. YOLOv7: Trainable bag-of-freebies sets new state-of-the-art for real-time object detectors. in 2023 IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), Jun. 2023, p. 7464–7475, doi: 10.1109/cvpr52729.2023.00721.
- [14] Li T and Zhao Q. AE-YOLO: An improved YOLOv7 based on attention enhancement for rail flaw detection. *IFAC-PapersOnLine.* 2023; 56(2): 7638–7643. doi: 10.1016/j.ifacol.2023.10.1162.
- [15] Ye G, Qu J, Tao J, Dai W, Mao Y and Jin Q. Autonomous surface crack identification of concrete structures based on the YOLOv7 algorithm. *Journal of Building Engineering.* 2023; 73: 106688. doi: 10.1016/j.jobe.2023.106688.
- [16] Xu G, Yue Q and Liu X. Real-time monitoring of concrete crack based on deep learning algorithms and image processing techniques. *Advanced Engineering Informatics.* 2023; 58: 102214. doi: 10.1016/j.aei.2023.102214.
- [17] Glenn J. v8.1.0 - YOLOv8 Oriented Bounding Boxes (OBB). 2024.
- [18] Xiong C, Zayed T and Abdelkader EM. A novel YOLOv8-GAM-Wise-IoU model for automated detection of bridge surface cracks. *Construction and Building Materials.* 2024; 414: 135025. doi: 10.1016/j.conbuildmat.2024.135025.